ENVIRONMENTAL VALUATION AND COST-BENEFIT ANALYSIS

This Chapter focuses on

- How can we calculate the cost and benefits of environmental outcomes?
- Can we put a value on ecological system, or on human life?
- How can we weight the interests of future generations?

GOODS

- Consumable item that is tangible in nature
- It possess value and satisfies human wants
- Classified into two:

MATERIAL GOODS	NON MATERIAL GOODS
• It can be seen, touched and transferred.	• It cannot be seen, touched and transferred.
• Example: cloth, car, shoes etc.	• Example: services rendered by Doctors, Teachers, Lawyers etc.

CLASSIFICATION OF GOODS – ON THE BASIS OF VALUE

FREE GOODS

- Free gift of nature
- Available without any limit
- Does not require human effort to produce
- Has no money value or price

ECONOMIC GOODS

- Not freely available
- They are not gifts of nature
- Requires human efforts for production
- Scarce in nature and has an attached value to it.

CLASSIFICATION OF GOODS – ON THE BASIS OF USE

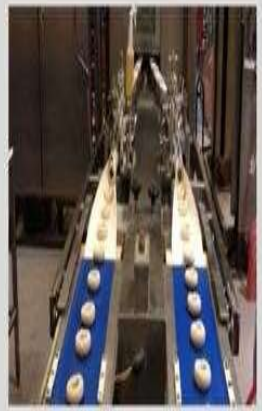
CONSUMER GOODS

- Direct or final consumption
- Purchased to satisfy current wants
- Three types of consumer goods

1. Durable goods
2. Non durable goods
3. Services

PRODUCER GOODS

- Used for production of other goods
- Machinery and other equipment used in the process
- Sold from one person to another
- Value added to all stages of production



CLASSIFICATION OF GOODS – ON THE BASIS OF OWNERSHIP

PRIVATE GOODS

- Manufactured and sold by private companies
- Entrepreneur is the holder of private goods
- Objective is to earn profit

PUBLIC GOODS

- Manufactured and sold by the government
- Government is the holder of public companies
- Objective is to bring overall growth and development

UTILITY

- Derived from the concept of usefulness
- Satisfying a consumer's want or need
- Economic utility influences the demand
- Value-in-use of a commodity

CHARACTERISTICS OF UTILITY

UTILITY IS SUBJECTIVE

Differs from person to person

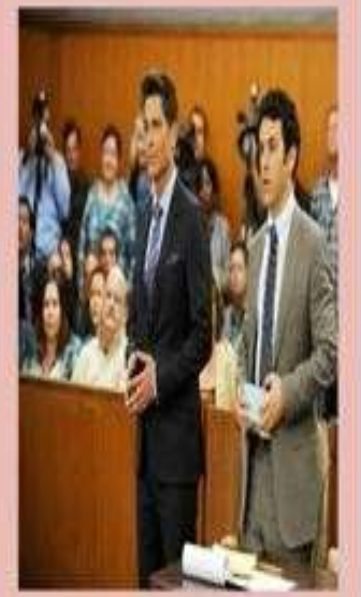
UTILITY IS RELATIVE

Different utility at different times or at different places

NO REAL EXISTENCE

Psychological factor

UTILITY IS ABSTRACT



Cannot be seen or touched

UTILITY IS NOT PLEASURE

Not necessary that commodity possessing utility may also give pleasure

Price
Value
Cost

COST
PRICE
VALUE

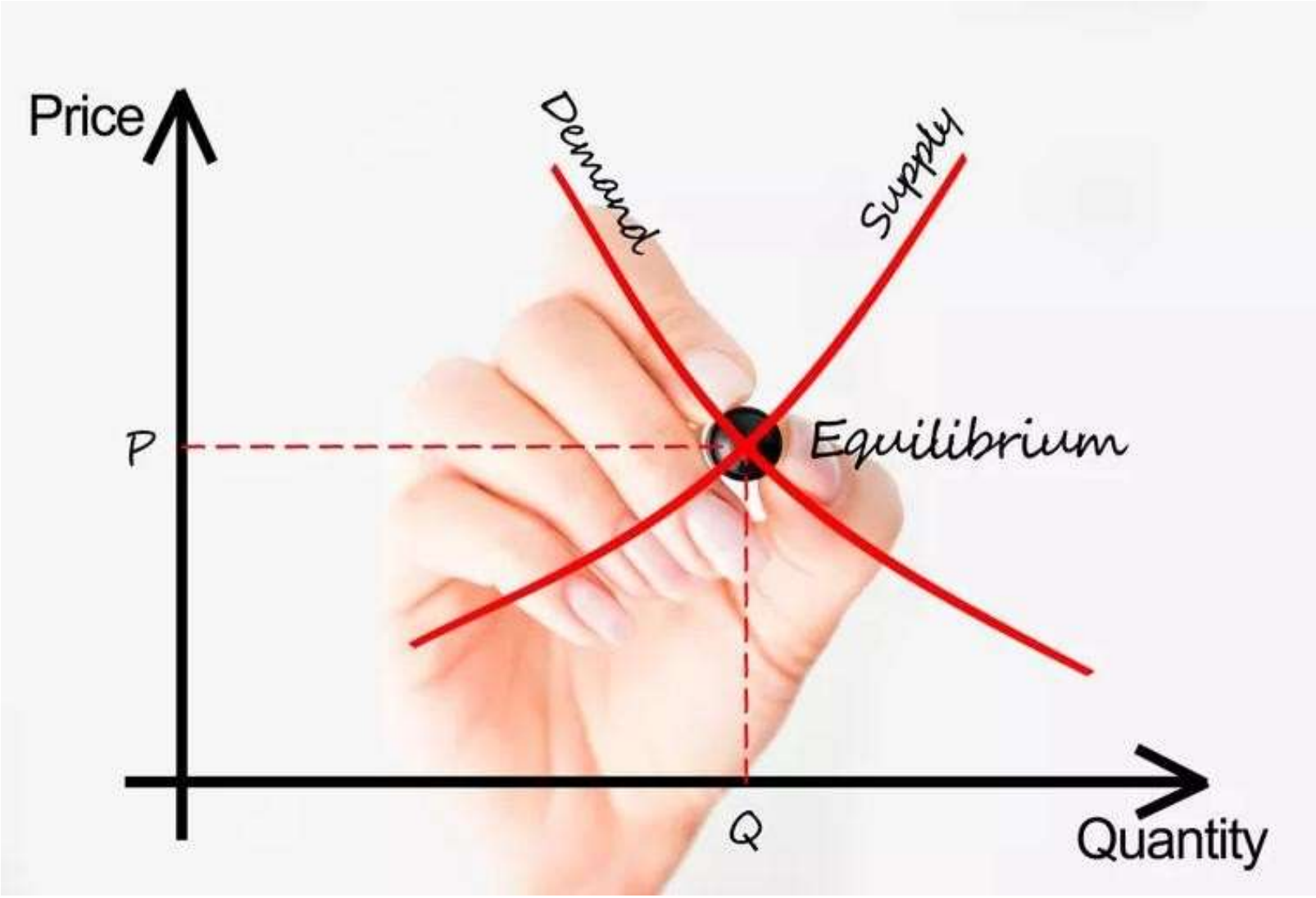
VALUE

- Value is the usefulness of any product to a customer.
- The value remains the same irrespective of the changes in the market.
- Example: spending 1000/- bucks a month on gym

PRICE

- Price is the amount of money paid by the buyer to the seller in exchange for any product and service.
- The Ups and downs in the market will affect the price.
- Example: Going to the mobile store to buy a phone

Private Goods.... *Recall* ..



CBA, benefits and costs are expressed in monetary terms and are adjusted for the time value of money, so that all flows of benefits and flows of project costs over time (which tend to occur at different points in time) are expressed on a common basis in terms of their net present value.

- The benefits and costs can be evaluated with respect to the world, a nation, a state, a county, a town, a city, a village, a neighborhood, a company, a group or an individual

- How do you quantify an increase in staff morale?
- What is the value of cleaner air or water?
- What is the value of a life that may be saved or lost?

An example
A controversy over dam construction

Suppose that there is a plan to construct a dam with deforestation.

Some people want the dam. But some people do not want the dam since they want to protect the forest.

Let's compare the cost and benefit of the dam construction.

cost

- Cost for purchasing the land
- Cost for purchasing the materials
- Cost for hiring people
- ・・・・etc.
- Cost for losing the forest

benefit

- Benefit of preventing flood
- ・・・・ etc.

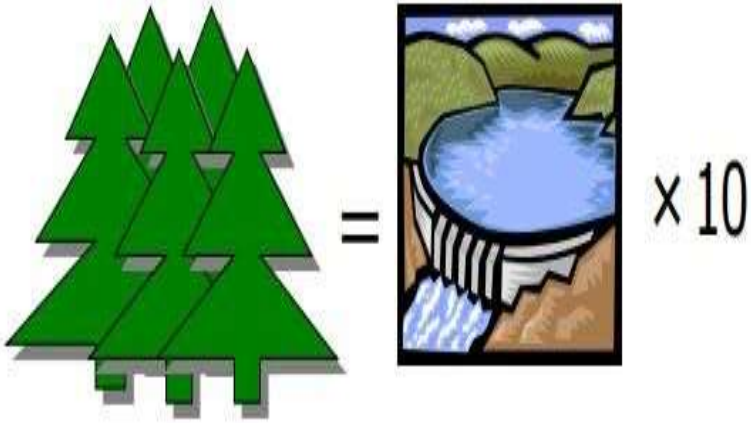
This is called the cost benefit analysis

What is the cost for losing the forest?

- While a forest gives people "happiness", certain aspects of it do not have a market price. This is because the aspects are not sold in the market.
  - Example: When we enjoy a beautiful view of the forest, we do not have to pay money unless the forest is part of a park where we have to pay an entrance fee.
- Because of the feature of the no market price, sometimes the aspect is dealt with as if it had no value. This is one of the causes of deforestation.
- Environmental economics tries to measure the value of these aspects and give a monetary value to the environment so that the value can be taken into consideration.
- How can we measure the monetary value of the environment?

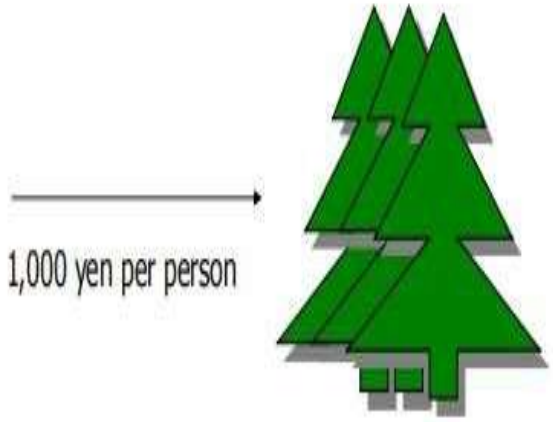
Example 1:
Giving monetary value to the water-retaining function of the forest.

How much does it cost to construct a dam which has the same amount of water-retaining function as the forest? Suppose the forest can retain 10 times the water that a dam can retain. Then 10 times the cost needed to construct a dam can be assumed to equal the monetary value of the water-retaining function of the forest.



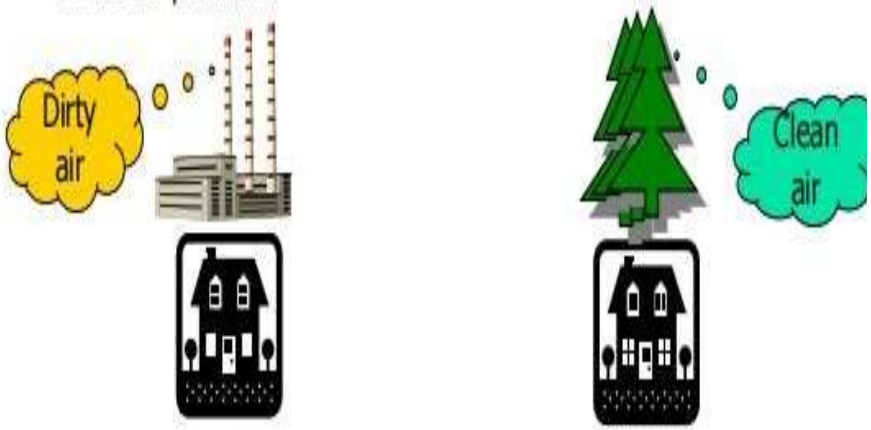
Example 2:
Giving monetary value to the recreation function of a forest.

How much are people willing to pay as the travel cost to go to the forest ? If they decide to go to the forest, they think that they will evaluate the satisfaction of going to the forest higher than the travel cost.



Example 3:
Giving monetary value to the clean air.

The price of a house is influenced by many factors. One of them is the degree of air pollution in the area. Suppose there are two houses and the characteristics of the houses are the same except for the air pollution. The size, layout, distance to the station, distance to the shops, and so on are the same about the two houses. But the degree of air pollution is different. In this case the difference of the house price can be considered to be the influence of the air pollution.

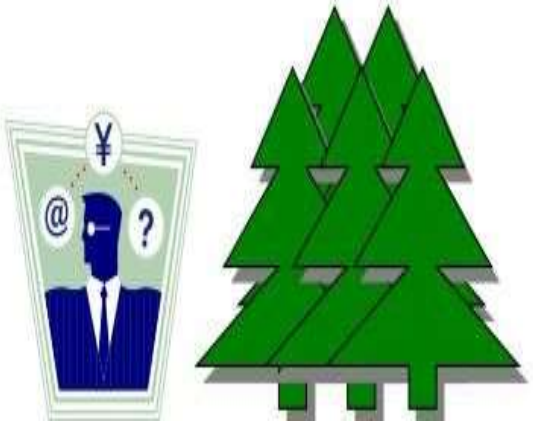


10 million yen

20 million yen

Example 4:
Giving monetary value to the function not related to the human use.

Examples 1, 2 and 3 are all related to human use. However, the forest has the function where it is not used by humans. In the valuation of these functions, price information cannot be used. Suppose an endangered animal lives in the forest. The forest provides food, water and so on to the animal. Humans do not use this habitat function. However, we may think this function is important. How do we give monetary value to the importance? In this case we can use a questionnaire to ask people how much they are willing to pay for the forest preservation for the animal.



Commercial Cost Benefit Analysis (CBA)

Total cost of the project

Expected future benefit from the project

- Benefit > Cost is desirable here.
- So it is nothing but a profitability analysis.
- But what will be the costs and/or the benefits that a society may have to bear and/or get from the proposed project are not considered here.

Non market benefits

Non mkt costs

$$\boxed{\text{NPV}} = \sum_{t=0}^{T} (B_t - C_t) \frac{1}{(1+r)^t}$$

Market vs social rates

We rank projects on the basis of NPV

We rely on consumer's WTP for public goods

$t[0, T]$

Issue: integrating non market factors into CBA

Why do we have to value the environment ?

- No price data exists for non- marketed environmental resources.
- Need information on benefits and costs of protecting and restoring the environment to guide budget allocations.
- Need information to guide compensation setting for victims.
- Need information to guide efficient and fair resource allocation.

Methods of Measuring Environmental Values

Revealed Preference :

Real choices of people are observed in some market and information are inferred on the trade-offs between money and the environmental good.

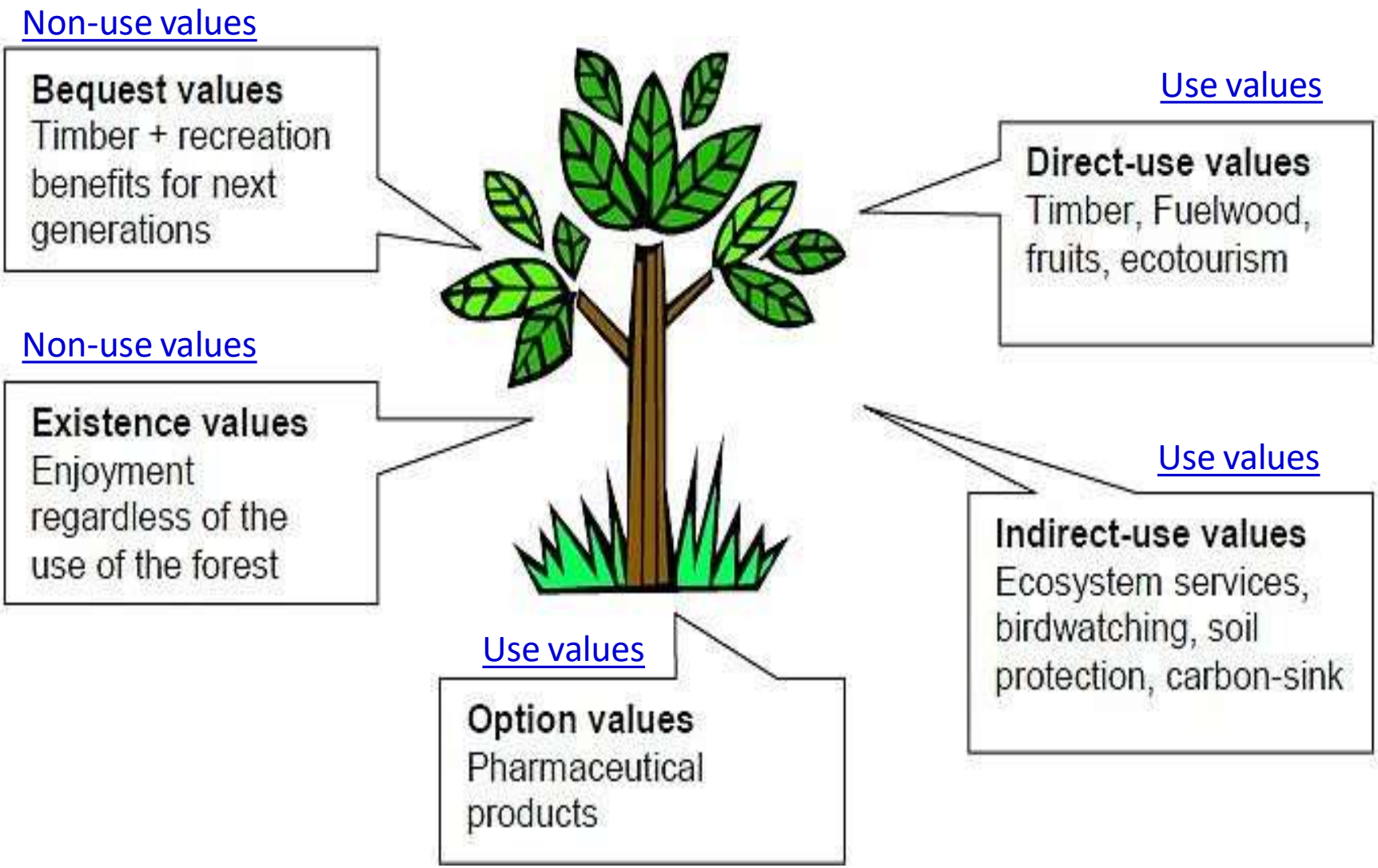
Stated Preference :

It basically involves asking people how much an environmental good is worth.

These information are collected through opinion polls or surveys.

Use Value		Non-Use Value		
1. Direct use value	2. Indirect use value	3. Option value	4. Bequest value	5.Existence value
Wood products (timber, fuel) Non-wood products(food, medicine, genetic material) Educational, recreational and cultural uses Human habitat	Watershed protection Nutrient cycling Air pollution reduction Micro-climatic regulation Carbon storage	Possible future uses of goods and services mentioned in 1 and 2 (use values) by actual stakeholders	Possible future uses of goods and services mentioned in 1 and 2 (use values) by the offspring of actual stakeholders	Bio-diversity Culture, heritage Benefits to stakeholders of only knowing of the existence of goods and services without using them
Valuation Tools				
Marketing analysis Related goods approach Travel cost method Contingent valuation method Hedonic pricing	Restoration cost Preventive expenditure Production function approach Replacement cost	Contingent valuation method	Contingent valuation method	Contingent valuation method

Figure: presents an example of the total economic value of a tropical forest.



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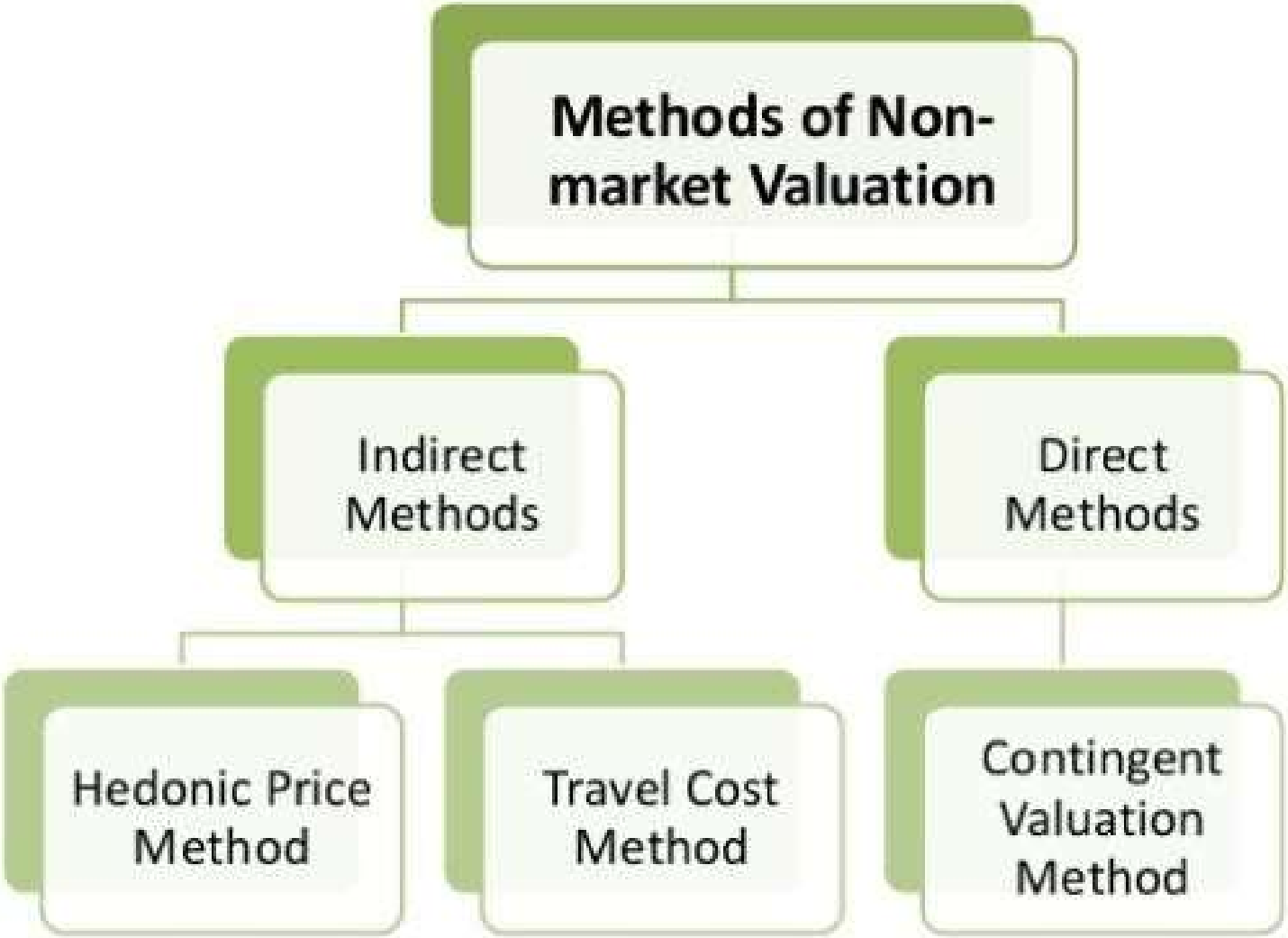
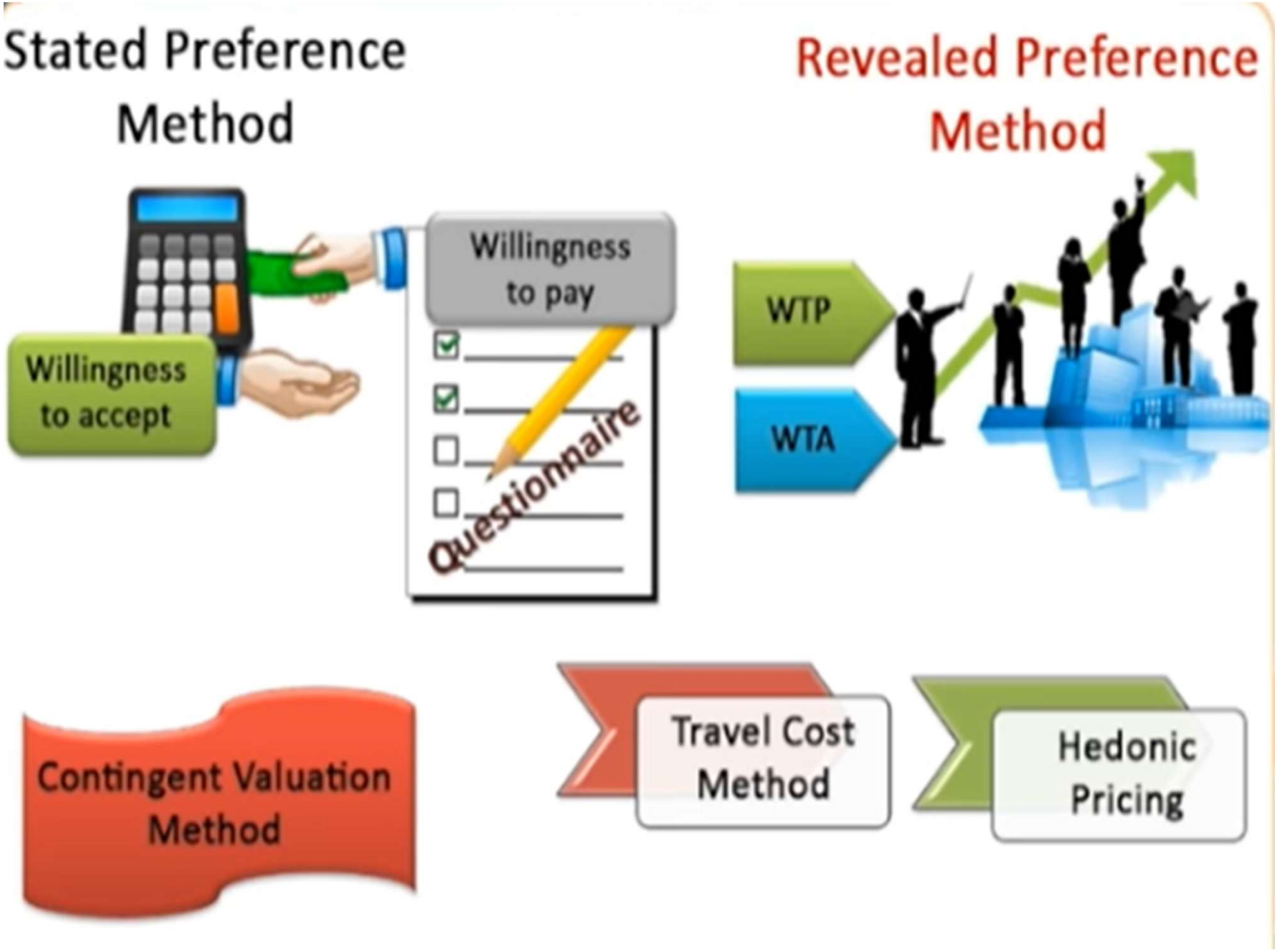

Methods of Non-market Valuation
Indirect Methods
Direct Methods
Hedonic Price Method
Travel Cost Method
Contingent Valuation Method


Stated Preference Method
Revealed Preference Method
Willingness to pay
Willingness to accept
Questionnaire
WTP
WTA
Contingent Valuation Method
Travel Cost Method
Hedonic Pricing

Hedonic Pricing Method

The hedonic pricing method is used to estimate economic values for ecosystem or environmental services that directly affect market prices. It is most commonly applied to variations in housing prices that reflect the value of local environmental attributes.

Hedonic Pricing Method

It can be used to estimate economic benefits or costs associated with:

- environmental quality, including air pollution, water pollution, or noise
- environmental amenities, such as aesthetic views or proximity to recreational sites

Hedonic Pricing

The use of statistical analysis to explain the price of a good or service as a function of several components.

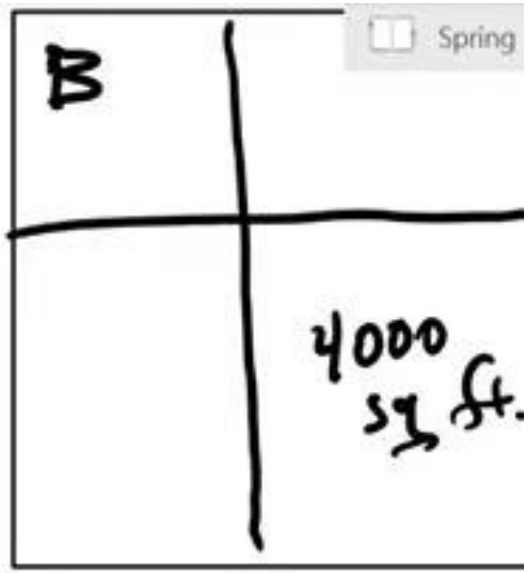
Example: hprice = f(#rooms, ...

What components helps explain the value of a house?

- Number of rooms – bedrooms | bathrooms
- Lot size (acreage)
- Age of house | style
- Local school quality
- Crime rates & other neighborhood traits
- Surrounding air quality or natural recreational opportunities

Schools A+
Crime : Low
Unemp : 5%

Pollution : High



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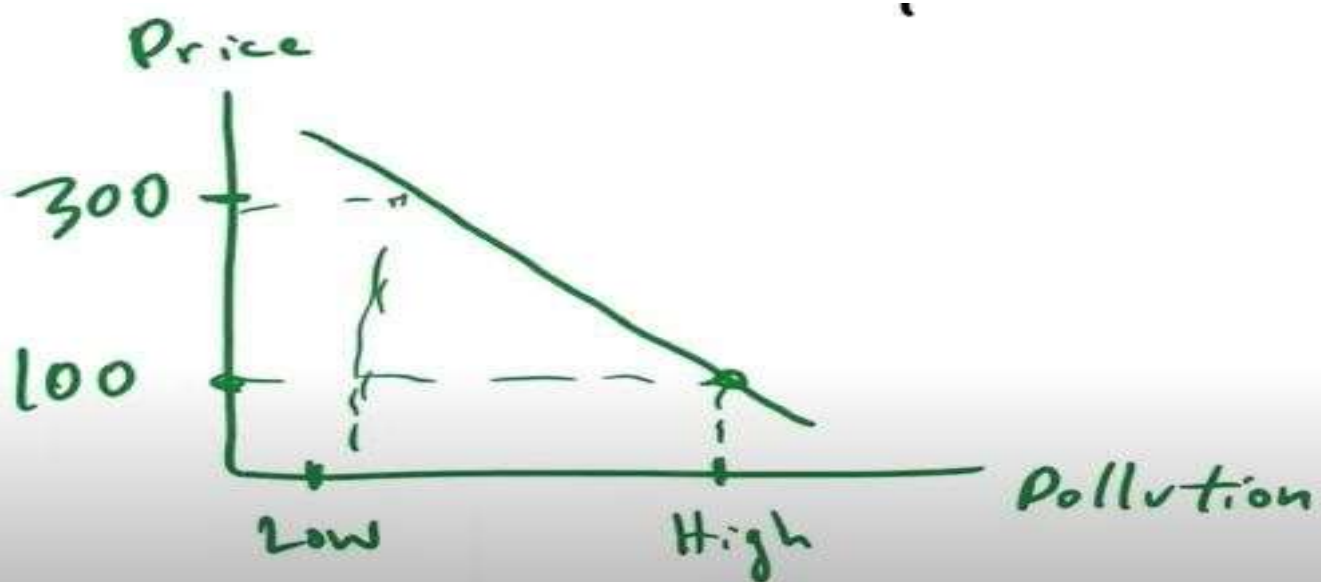


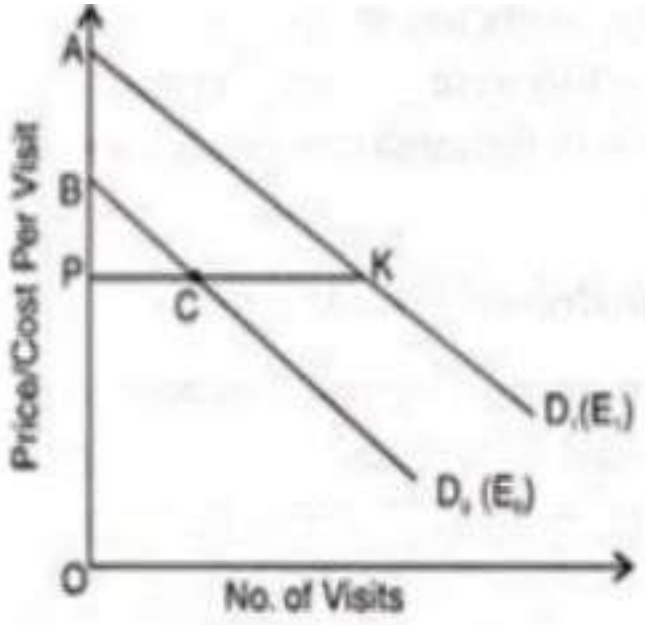
Examples:

- (Kling et. Al., 2015): Proximity to a national or state park increased property values in Northern Colorado, while proximity to a national forest has no significant impact.
- Apartment sale prices in Vienna, Austria increased the closer they were to the city's greenbelt.
- Homes located near hazardous waste sites, airports, and highways are associated with lower prices.

Indirect valuation techniques

The travel cost method is an indirect valuation technique that was designed to model recreation behavior.

This method calculates a value based on the fact that <u>the price paid to travel to the site is the ultimate value of that site</u>.



The travel cost method

The costs associated with travelling to the resource (fuel, mechanical maintenance of vehicle, time spent travelling there) become the variables to be used to determine the value of a resource.

The weakness of this method is that, it only deals with single destination trips and assumes that travel is a means, rather than an end in itself

Stated Preference Methods

Economic valuation methods based on survey responses to hypothetical scenarios.

Contingent Valuation

Contingent Valuation (CV) is an economic tool that uses surveys to question people regarding their WTP for a good.

- Sometimes, CV studies use Willingness to Accept (WTA) instead of WTP estimates
- Example:
  - "What is the minimum amount of compensation you are WTA for a 10 decrease in air quality?"
  - "What is the maximum amount you are WTP for a 10 percent increase in air quality?"

In economic theory, the willingness to pay and willingness to accept values are similar, but in reality it has been demonstrated beyond doubt that willingness to accept values can be four times higher than willingness to pay.

WHY CBA??

- To determine if an investment decision is sound, calculating if its benefits outweigh its costs and by how much
- To provide a basis for comparing alternative investment options, comparing the total expected cost of each option with its total expected benefits, thereby creating a basis for selecting the most desirable/viable option

A simple definition of a CBA

- The regulator should choose the project with largest positive expected net present value: $E[NPV] > 0$
- **Net benefits**, at time t:

$$NB_t = B(x) - c(x)$$
$$= \sum_{i=1}^{n} WTP_t - c(x)$$

The sum of what all relevant agents want to pay, minus the costs

- **Net present value**:

$$NPV = \sum_{t=1}^{T} \beta^t \, NB_t$$

The sum of all discounted net benefits over a project's time horizon.

Choice of discount rate

There is disagreement about the discount rate that should be used in CBA

This matters because the result of the NPV test can be very sensitive to the number used for the discount rate

$$NPV = \sum_{t=1}^{T} \left(\frac{1}{1+r}\right)^t NB_t$$

Especially if the project *lifetime* is long (often with environmental consequences)

Non market benefits

Non mkt costs

$$\boxed{\text{NPV}} = \sum_{t=0}^{T} (B_t - C_t) \frac{1}{(1+r)^t}$$

Market vs social rates

We rank projects on the basis of NPV

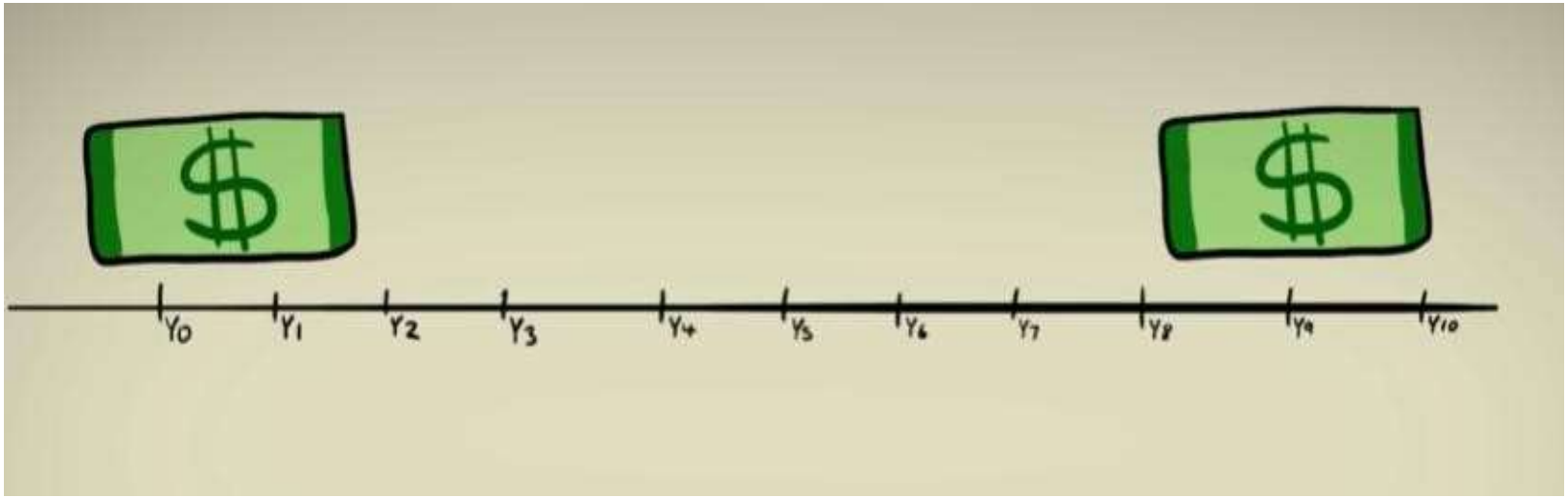
We rely on consumer's WTP for public goods

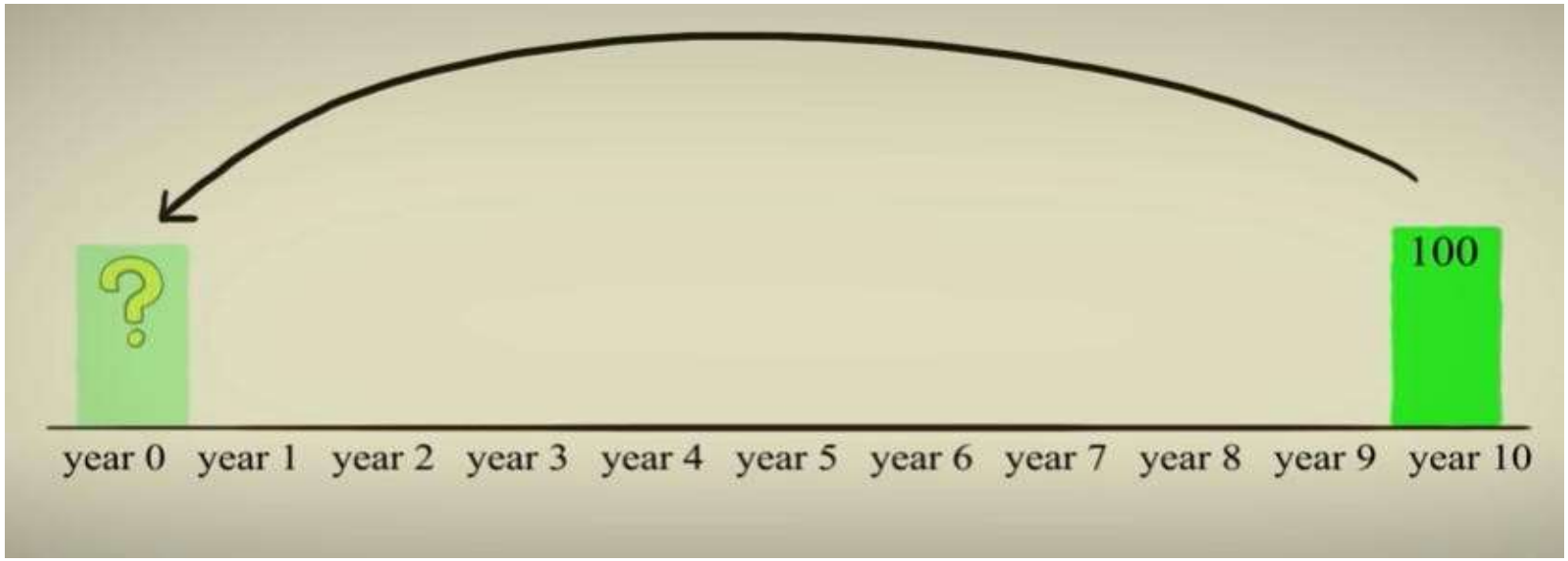
$t[0, T]$

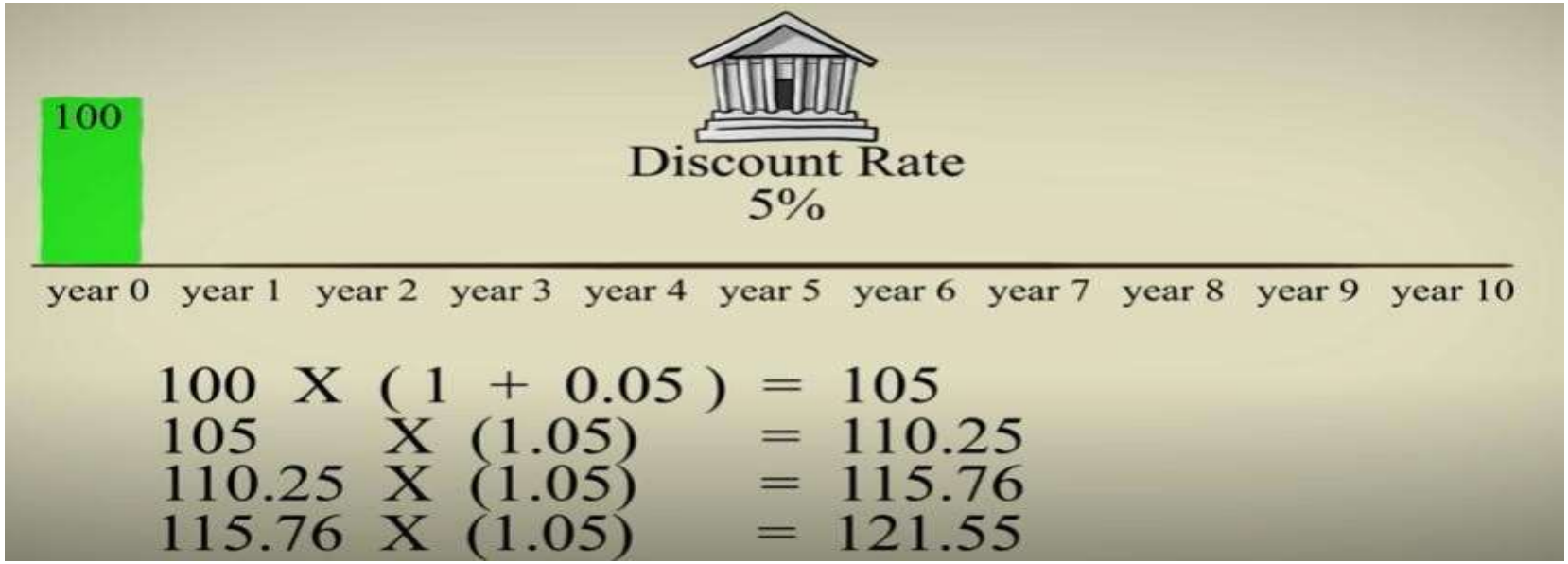
Issue: integrating non market factors into CBA

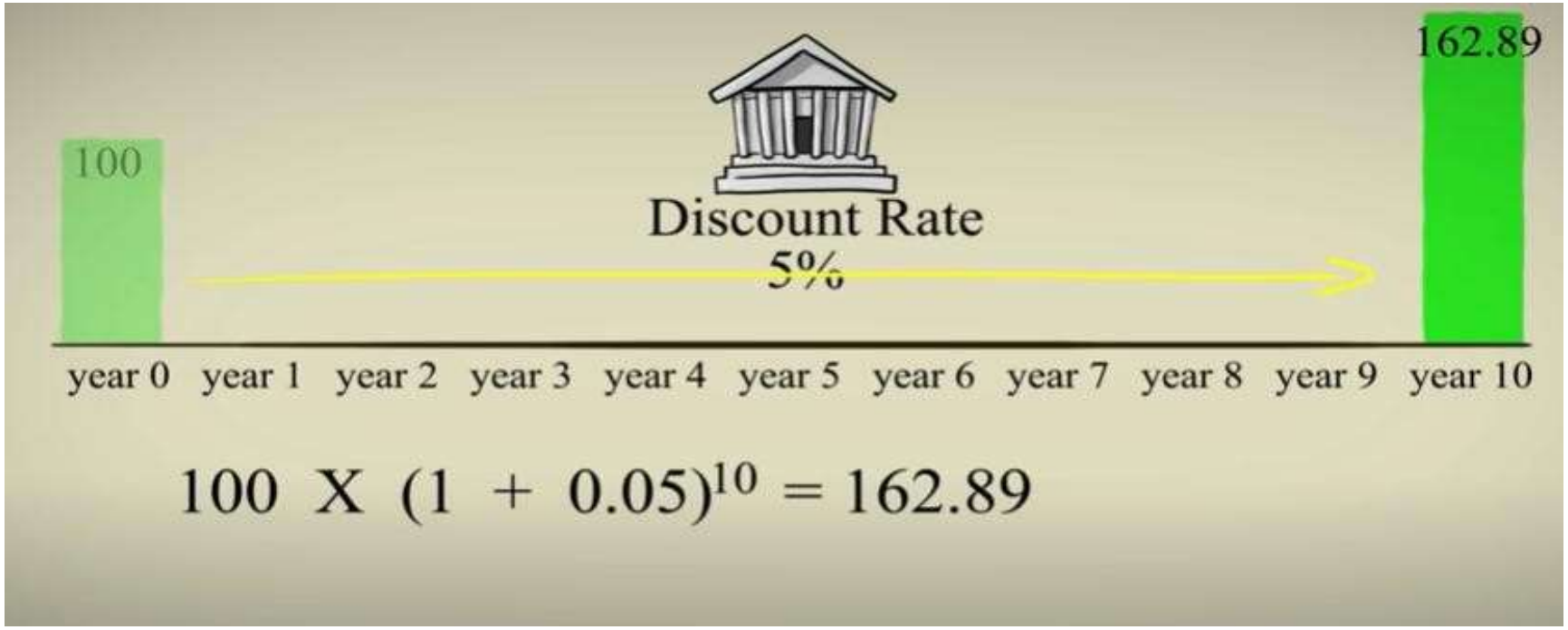
The Rationale for Discounting

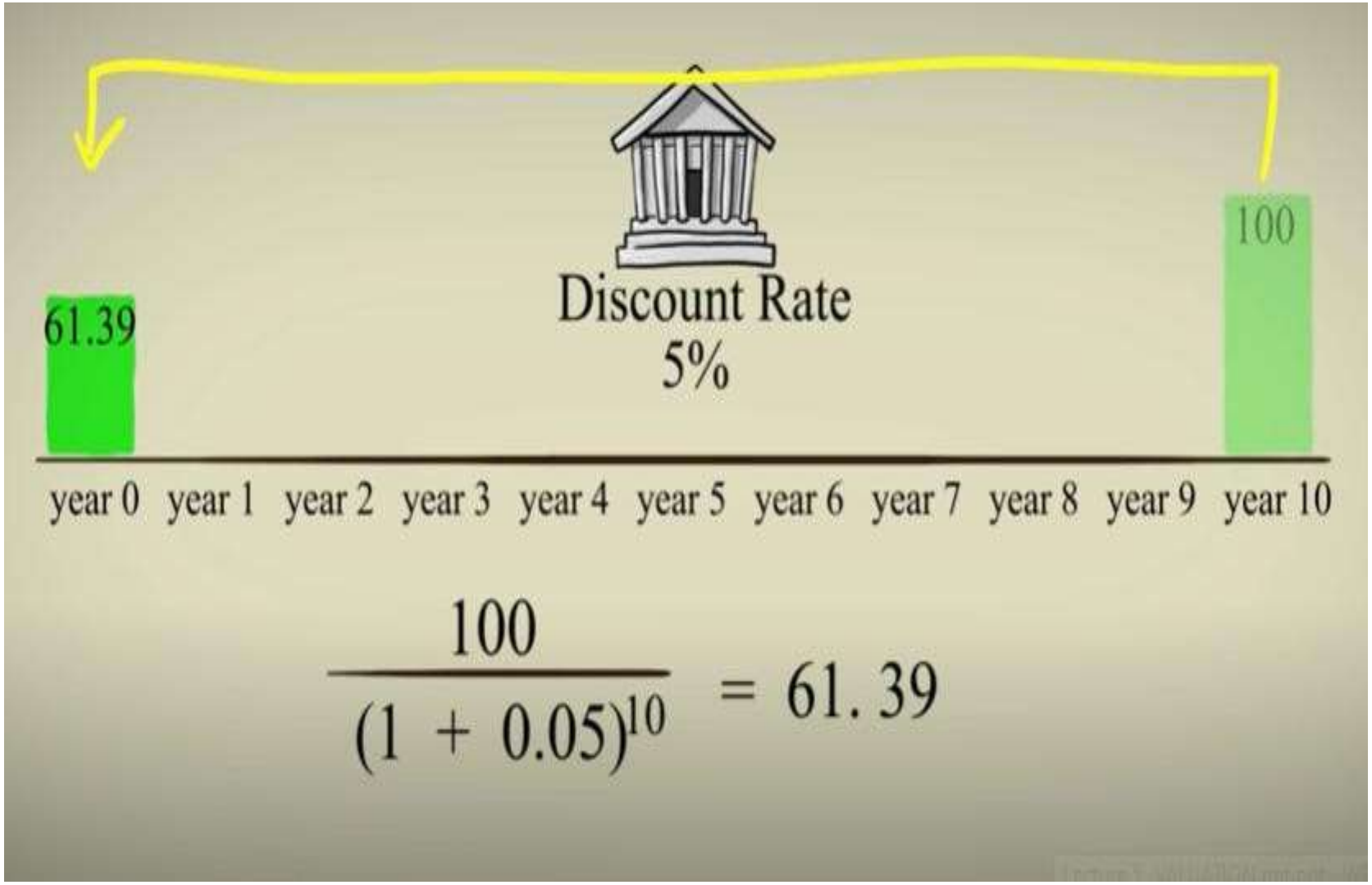
- Inflation
- Opportunity cost of capital
- Public projects involve uncertainty and risk
- Humans prefer near-term benefits to those that accrue in the distant future

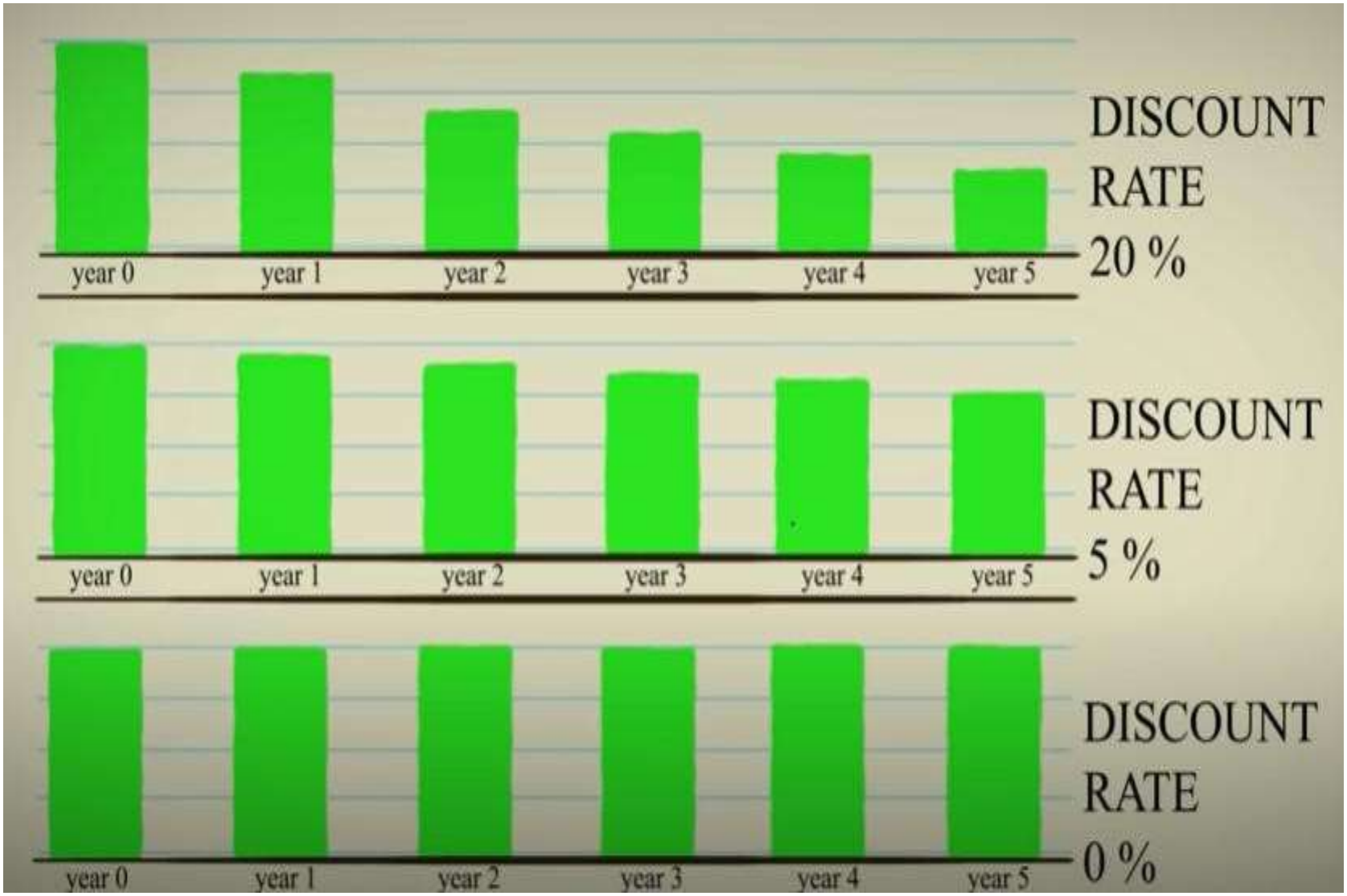

$
$
Y0
Y1
Y2
Y3
Y4
Y5
Y6
Y7
Y8
Y9
Y10


?
100
year 0
year 1
year 2
year 3
year 4
year 5
year 6
year 7
year 8
year 9
year 10


100
Discount Rate
5%
year 0 year 1 year 2 year 3 year 4 year 5 year 6 year 7 year 8 year 9 year 10
100 X (1 + 0.05) = 105
105 X (1.05) = 110.25
110.25 X (1.05) = 115.76
115.76 X (1.05) = 121.55


162.89
100
Discount Rate
5%
year 0 year 1 year 2 year 3 year 4 year 5 year 6 year 7 year 8 year 9 year 10
100 X (1 + 0.05)10 = 162.89


61.39
Discount Rate
5%
100
year 0
year 1
year 2
year 3
year 4
year 5
year 6
year 7
year 8
year 9
year 10
$$\frac{100}{(1 + 0.05)^{10}} = 61.39$$


year 0
year 1
year 2
year 3
year 4
year 5
DISCOUNT RATE 20 %
year 0
year 1
year 2
year 3
year 4
year 5
DISCOUNT RATE 5 %
year 0
year 1
year 2
year 3
year 4
year 5
DISCOUNT RATE 0 %

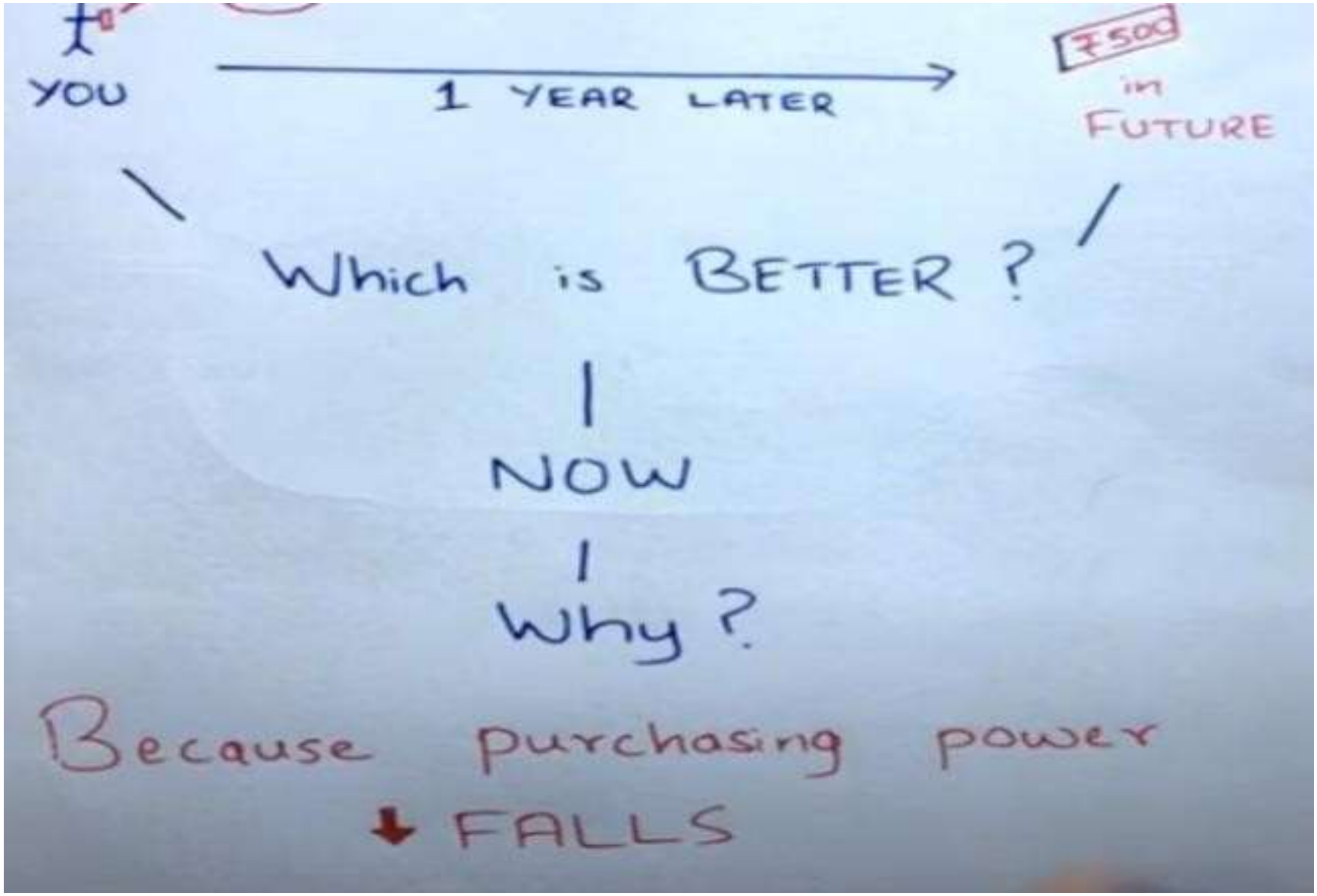
- Discount rates can vary from 0 to infinity
- Between these two extremes, we typically consider discount rates from 1 to 30 percent
- Most environmentally based analyses or inter generational policies should use discount rates that are closer to 2 to 5%
- Businesses tend to use discount rates in the 10 to 20% range in order to justify investing in the current analysis as opposed to other opportunities

#4 INVESTMENT DECISION

NET PRESENT VALUE

The NPV technique is a discounted cash flow method that considers time value of money in evaluating capital investments.

It is a method of calculating the present value of cash flows (inflows & outflows) of an investment proposal using the cost of capital as an appropriate discounting rate.


YOU
1 YEAR LATER
₹500
in
FUTURE
Which is BETTER ?
NOW
Why ?
Because purchasing power
FALLS

DISCOUNTING

PV ← FV

PRESENT VALUE

FUTURE VALUE

₹455 PV ← 10% ₹500 A YEAR LATER

$$PV = FV \frac{1}{(1+r)^n}$$

r = interest rate

n = no. of years

$$= 500 \times \frac{1}{1.10}$$

$$= 455$$

FORMAT

YEAR	CASH FLOWS	DISCOUNTING FACTOR	PRESENT VALUE
0	(Initial Investment)		
1			
2			
3			
		NPV	X X X X

Decision Rule

If NPV > 0 ACCEPT ✓

If NPV < 0 REJECT ✗

1. Calculate NPV of the two projects and suggest which of the two projects should be accepted assuming discount rate @ 10%

	Project X	Project Y
Initial Investment	₹40,000	₹60,000
Estimated Life	5yrs	5yrs
Scrap Value	₹2,000	₹4,000

Cash Inflows

YEAR	1	2	3	4	5
PROJECT X	10,000	20,000	20,000	6,000	4,000
PROJECT Y	40,000	20,000	10,000	6,000	4,000

PV Factor @ 10% Discount

YEAR	1	2	3	4	5
PV Factors @10%	0.909	0.826	0.751	0.683	0.621

PROJECT X

YEAR	CASHFLOWS	DF @ 10%	PV
0	(40,000)	1	(40,000)
1	10,000	0.909	9,090
2	20,000	0.826	16,520
3	20,000	0.751	15,020
4	6,000	0.683	4,098
5	4,000	0.621	2,484
5 (Scrap)	2,000	0.621	1,242
		NPV	8,454

PROJECT Y

YEAR	CASHFLOWS	DF @ 10%	PV
0	(60,000)	1	(60,000)
1	40,000	0.909	36,360
2	20,000	0.826	16,520
3	10,000	0.751	7,510
4	6,000	0.683	4,098
5	4,000	0.621	2,484
5 Scrap)	4,000	0.621	2,484
		NPV	9,456

Thank You
&
Happy Learning!